

## Ph.D. Algebra Exam – May 12, 2009

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Answer exactly **seven** of the following eleven problems.

You must show your work. Answers with no work and/or no explanations will receive no credit. State clearly any theorem you use in your proofs.

In the problems,  $\mathbf{Q}$  and  $\mathbf{C}$  are the fields of rational numbers and complex numbers respectively

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1. Let  $\mathcal{C}$  be a category, and denote by **Set** the category of sets. (i) Define what it means for a functor  $F : \mathcal{C}^{op} \rightarrow \mathbf{Set}$  to be representable. (ii) Show that if  $X, Y$  are objects of  $\mathcal{C}$  representing the same functor  $F : \mathcal{C}^{op} \rightarrow \mathbf{Set}$ , then  $X \simeq Y$ .

2. Let  $R$  be a ring with identity. Show that the following are equivalent: (i) Every unitary left  $R$ -module is projective, (ii) every unitary left  $R$ -module is injective, (iii) every exact sequence of unitary left  $R$ -modules splits.

3. If  $G$  is a group, denote by  $G^{ab}$  the abelianization of  $G$ , and if  $X$  is a set, denote by  $F(X)$  and  $F^{ab}(X)$  the free group on  $X$  and the free abelian group on  $X$  respectively. Show that  $F(X)^{ab} \simeq F^{ab}(X)$ .

4. Let  $K$  be a field,  $L = K(\alpha)$  a simple extension of  $K$ , and  $\overline{K}$  an algebraic closure of  $K$ . Show that  $\alpha$  is separable over  $K$  if and only if the ring  $L \otimes_K \overline{K}$  has no nilpotent elements.

5. Show that if  $R$  is a Noetherian commutative ring with identity, then  $R[X]$  is also Noetherian.

6. State and prove Artin's theorem on the linear independence of characters. and use it to prove the multiplicative form of Hilbert's Theorem 90.

7. Let  $p(x) = x^7 - 3 \in \mathbf{Q}[x]$ ,  $\zeta \in \mathbf{C}$  a primitive 7-th root of 1, and  $K$  the splitting field of  $p(x)$ . (i) Compute the Galois group of  $K/\mathbf{Q}$ . (ii) Compute the Galois group of  $K/\mathbf{Q}(\zeta)$ .

8. (i) Give an example of extensions  $E/L$ ,  $L/K$  where  $E/K$  is normal and  $L/K$  is not. (ii) Give an example of normal extensions  $L/K$ ,  $E/L$  such that  $E/K$  is not normal.

9. Let  $R$  be a Dedekind domain, and let  $I$  be a non-zero ideal of  $R$ . (i) Show that  $R/I$  is Artinian. (ii) Show that if a Dedekind domain is Artinian then it is a field.

10. Show that if  $L/K$  is any extension of fields, there is a algebraically independent set  $\{\alpha_i\}_{i \in I}$  of elements of  $L$  such that  $L$  is algebraic over  $K(\alpha_i, i \in I)$ .

11. Let  $k$  be a field. A  $k$ -algebra  $R$  is *central* if the image of the structure map  $k \rightarrow R$  is in the center of  $R$ . (i) Suppose  $D$  is a central  $k$ -algebra that is a division algebra (in particular,  $D$  is a  $k$ -vector space). Show that if  $k$  is algebraically closed and  $D$  has finite dimension as a  $k$ -vector space, then  $D = k$ .

(ii) Use this result to show that if  $k$  is algebraically closed and  $R$  is a central  $k$ -algebra that is semisimple and of finite dimension over  $k$ , then  $R$  is a product of matrix algebras over  $k$ .